

WildEcoWatch : An Integrated IoT and AI Framework for Wildlife Surveillance and Conservation

Anuj Gavhane
*Electronics & Telecommunication
Pimpri Chinchwad College of
Engineering
Pune, India*
anuj.gavhane21@pccoepune.org

Abhishek Balgude
*Electronics & Telecommunication
Pimpri Chinchwad College of
Engineering
Pune, India*
abhishek.balgude21@pccoepune.org

Sushant Awate
*Electronics & Telecommunication
Pimpri Chinchwad College of
Engineering
Pune, India*
sushant.awate21@pccoepune.org

Veena Kulkarni
*Electronics & Telecommunication
Pimpri Chinchwad College of
Engineering
Pune, India*
veena.kulkarni@pccoepune.org

Abstract— The WildEcoWatch project addresses the critical issue of wildlife crimes, such as poaching and illegal activities, which threaten global biodiversity and ecosystems. This research proposes an integrated IoT and AI framework designed to enhance wildlife surveillance and conservation efforts. By leveraging IoT sensors, machine learning algorithms, and real-time alert mechanisms, the system monitors forests for unauthorized intrusions and disturbances. The framework utilizes Raspberry Pi-based sensors and advanced machine learning models to detect and classify human and wildlife activities, sending instant alerts via SMS and Text-to-Speech (TTS) notifications to forest rangers and conservationists. Augmented Reality (AR) is also integrated to provide real-time visual information to field personnel, enabling swift decision-making. The system is designed to be cost-effective, scalable, and operable in remote areas with minimal infrastructure. WildEcoWatch aims to provide a sustainable solution for wildlife protection, poaching prevention, and global conservation efforts.

Keywords—*Artificial Intelligence, Augmented Reality, Environmental Protection, Internet of Things (IoT), Machine Learning, Poaching Detection, Raspberry Pi, Real-Time Monitoring, and Wildlife Conservation.*

I. INTRODUCTION

Wildlife conservation is a critical component of maintaining global biodiversity and ecological balance. However, human activities such as poaching, illegal logging, and habitat destruction pose significant threats to forests and endangered species. Traditional methods of wildlife monitoring, such as manual patrols and basic camera traps, are often inefficient, time-consuming, and unable to provide real-time responses to threats. These limitations leave ecosystems vulnerable to illegal activities, making it challenging to protect wildlife and maintain environmental stability.

To address the challenges, the WildEcoWatch project proposes an innovative solution that integrates modern technologies such as the Internet of Things (IoT), machine learning, and embedded systems. The system is designed to provide real-time monitoring and instant alerts to authorities when intrusions or disturbances are detected in forest areas. By deploying IoT devices such as cameras, motion sensors, and Raspberry Pi-based systems, WildEcoWatch enables continuous surveillance and rapid response to potential threats. The project falls under the domain of environmental conservation and embedded systems, aiming to bridge the gap between technology and wildlife protection.

The primary motivation behind this project is the growing need for an efficient and cost-effective system to combat wildlife crimes, particularly in remote and resource-constrained regions. Current surveillance systems often lack the capability to detect intrusions in real-time, leaving wildlife at risk. WildEcoWatch seeks to overcome these limitations by leveraging IoT and machine learning technologies to detect unauthorized activities and send instant alerts to forest rangers, conservationists, and law enforcement agencies. This proactive approach not only enhances the ability to protect endangered species but also contributes to the preservation of ecological balance.

The objectives of the WildEcoWatch project include the development of an intelligent surveillance system capable of detecting poaching and other disruptions in rainforests. The system utilizes pre-trained machine learning models to identify intrusions and wildlife disturbances, providing real-time alerts through SMS and notifications. Additionally, the project incorporates Augmented Reality (AR) to assist field personnel with real-time visual information, improving decision-making during critical situations. The system is designed to be scalable, affordable, and adaptable to diverse ecosystems, making it suitable for deployment in remote areas with minimal infrastructure.

In conclusion, WildEcoWatch represents a significant step forward in wildlife conservation by combining IoT, AI, and real-time monitoring technologies. The project aims to provide a sustainable and effective solution for protecting wildlife, preventing poaching, and contributing to global conservation efforts. By addressing the limitations of traditional surveillance methods, WildEcoWatch has the potential to revolutionize wildlife monitoring and management practices, ensuring the protection of biodiversity for future generations.

II. LITERATURE REVIEW

A wildlife intrusion detection system using YOLOv5 was proposed to monitor farming areas and detect animals such as elephants, tigers, and wild boars. The model was trained on 280 images and achieved a mean average precision (mAP) of 94%, demonstrating high accuracy in real-time detection. This system significantly improved wildlife intrusion monitoring while reducing false alarms [1]. An improved YOLOv8-based detection system was developed to identify four animal species: buffalo, elephant, rhino, and zebra. The study utilized mosaic data augmentation and advanced convolutional layers, achieving an mAP of 99.4% on training data and 95.9% on validation data. This approach increased detection efficiency, making it suitable for real-time wildlife monitoring [2]. An ESP32 camera-based IoT intrusion detection model was designed to monitor forests using ultrasonic sensors, an NRF24L01 transmitter, and an Arduino Uno. The system activates an E-vehicle upon intrusion detection and captures images for further analysis. This model provides an affordable solution for real-time tracking of wild animals, improving ecosystem protection strategies [3].

A farm security enhancement system using YOLOv3 and OpenCV was proposed for real-time monitoring of agricultural areas. It classified animals into harmful and harmless categories and triggered alarms for dangerous intruders. The system effectively improved crop management, early detection of animal intrusions, and false alarm reduction [4]. A deep learning-based animal intrusion detection model using YOLO and intelligent surveillance was developed for crop protection. The system included feature extraction, pre-processing, and image classification and sent real-time alerts using a GSM module. The findings showed that automated image processing significantly enhanced wildlife monitoring while ensuring low-cost and energy-efficient deployment [5].

An AI-powered real-time animal detection system using YOLOv7 on Raspberry Pi was implemented to analyze live video feeds and detect animal intrusions. Upon detection, the system triggered specific repellent sounds and sent SMS alerts to farmers. This approach demonstrated high accuracy and improved response times, making it a viable solution for agricultural protection [6]. A Fast R-CNN-based intrusion detection system was designed to integrate Raspberry Pi, thermal sensors, and motion detectors for wildlife monitoring. The study compared Fast R-CNN with YOLO and SSD models, demonstrating superior accuracy and reduced processing time for wildlife tracking and conservation efforts [7].

A GSM-based animal detection system was implemented using ultrasonic sensors and Arduino Uno to monitor farm intrusions. The system triggered alarms and sent alerts to farmers when animals were detected. This approach provided 24/7 real-time monitoring, effectively reducing financial losses and improving farm security [8]. A YOLOv5 and YOLOv8-based animal detection model was designed for video-based surveillance systems. It was trained on the COCO dataset and implemented a user interface for live image input. The results showed that YOLOv8 achieved 85% accuracy, outperforming YOLOv5 in precision but requiring more computational resources [9]. An automated wild animal detection and alert system using YOLOv5 was proposed for community safety. The system utilized video feeds, sound deterrents, and mobile notifications to prevent human-wildlife conflicts. Future improvements focused on enhancing accuracy and expanding the system to diverse wildlife regions [10].

A deep learning-based animal classification system was developed using depth-wise separable convolution layers and zero padding for image processing. The system was tested on the IWildCam dataset, achieving an accuracy of 99.6% and an intersection over union (IOU) score of 0.878, demonstrating its effectiveness in wildlife identification [11]. An Arduino-controlled crop protection system was introduced to safeguard farmlands from wild animals. It used PIR motion sensors, buzzers, LED lights, and GSM modules for alert mechanisms. This system was cost-effective and improved farmers' ability to protect their crops without harming animals [12].

A real-time IoT-based animal intrusion prevention system was implemented to prevent wildlife conflicts in agricultural lands. It integrated machine learning techniques, ultrasonic sound deterrents, and SMS/email notifications to ensure efficient wildlife tracking and farm protection [13]. An AIoT-based intrusion detection system was designed using YOLOv3, PIR motion sensors, and Raspberry Pi. The system automatically captured images, activated alarms, and triggered emergency alerts, enhancing wildlife conservation efforts and minimizing human-wildlife conflicts [14]. A YOLOv4 and LoRa-based animal detection system was introduced for wildlife tracking. It utilized Raspberry Pi, ESP32 communication, and K-means clustering to predict animal movement patterns. This system provided timely alerts to forest authorities, reducing property damage and improving conservation strategies [15].

III. METHODOLOGY

The WildEcoWatch project leverages extensive research to develop an advanced wildlife monitoring system. Studies emphasize the importance of IoT, AI, and sensor networks in tackling issues like wildlife poaching, crop damage, and human-wildlife conflicts. AI-driven solutions, including machine learning models and automated alert systems, enhance real-time threat detection and intrusion prevention. By integrating these technologies, WildEcoWatch ensures high accuracy while reducing false alarms, making it a scalable and reliable solution across various environments. A key feature of WildEcoWatch is its seamless integration of hardware and software for real-time performance. The YOLO object detection model, known for its speed and precision,

enables fast animal identification, while PIR motion sensors enhance surveillance by detecting movement. With cloud-based storage and remote access, the system provides continuous tracking and analysis. Designed for precision, scalability, and efficiency, WildEcoWatch adapts to diverse terrains, offering a sustainable and effective solution for wildlife conservation.



Figure 1- Block Diagram

The Animal Intrusion Detection System consists of multiple components working together to monitor and deter wildlife threats. A real-time video camera continuously captures footage, while motion and infrared (IR) sensors detect movement and ensure visibility in low-light conditions. These inputs are processed by an AI-based object detection model, which identifies animals and distinguishes them from other objects.

A microcontroller or edge device acts as the system's control hub, integrating sensor inputs, AI detections, and classification results to trigger appropriate actions. If a threat is detected, the system sends real-time notifications via mobile apps, web dashboards, or emails. Additionally, a repellent system using sound or light deterrents is activated to ward off dangerous animals. All events are logged and stored, including video footage and sensor data, for future analysis, performance evaluation, and ecological monitoring.

The flowchart [2] outlines the detailed operational workflow of the Animal Intrusion Detection System, illustrating the step-by-step process of detecting and responding to potential wildlife threats. The system starts with initialization, where all components, including sensors, the microcontroller, and AI models, are prepared for monitoring. The Passive Infrared (PIR) sensor is then activated to detect motion by sensing changes in infrared radiation within the defined surveillance area. The system continuously monitors for movement, and if no motion is detected it remains in a standby mode, ensuring energy efficiency while maintaining constant vigilance

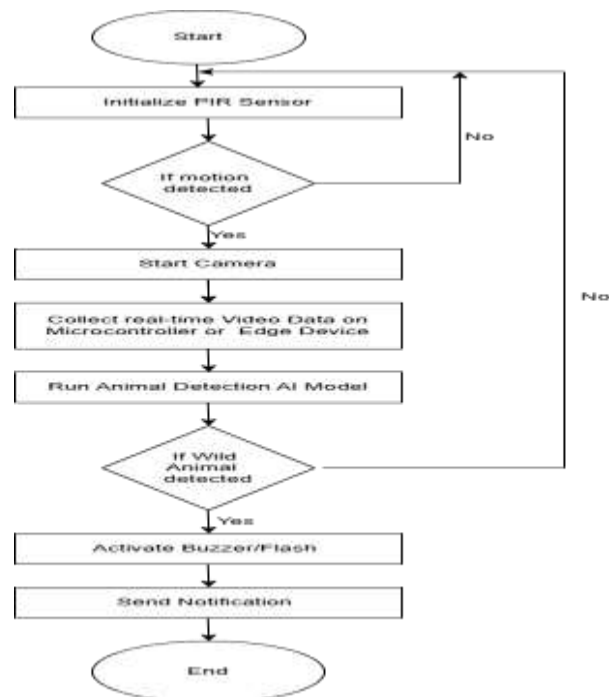


Figure 2- Flowchart

Upon motion detection, the camera is activated to capture real-time video footage of the monitoring area. The recorded video data is then transmitted to a microcontroller or edge device, such as a Raspberry Pi or Jetson Nano, which serves as the control hub for processing sensor inputs and executing decision-making algorithms. The captured footage undergoes AI-based object detection, where an advanced deep learning model is applied to identify the presence of animals and differentiate them from other objects or human activity. The AI model then classifies the detected object to determine whether it is a wild animal, which poses a potential threat to farms or residential areas.

If the detected object is identified as a wild animal, the system automatically triggers a set of deterrent mechanisms to prevent further intrusion. These deterrents include buzzers and flashing lights, which serve as non-lethal methods to repel the animal without causing harm. Additionally, a real-time notification alert is sent to the user via various communication channels, such as a mobile application, web dashboard, or SMS, ensuring that the concerned individuals or authorities are informed of the detected threat. The system then continues to operate, maintaining continuous monitoring and logging event data, including video footage and sensor readings, for future analysis, performance evaluation, and ecological monitoring. By integrating real-time detection, AI-based classification, and automated deterrents, the system provides an efficient, proactive, and data-driven approach to wildlife intrusion management.

Software implementation : The software implementation of the Animal Intrusion Detection System focuses on developing an efficient pipeline for real-time animal detection and classification. For the initial phase, the system takes an input image from the user and processes it. The frontend interface allows users to upload an image, which is

then analyzed by the backend AI model to determine the presence of an animal. The model predicts the detected animal along with a confidence score, which is displayed on the frontend to indicate the accuracy of the detection

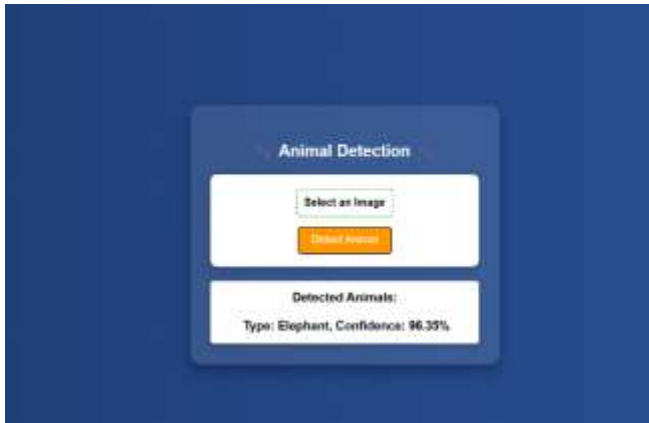


Figure 3- frontend interface

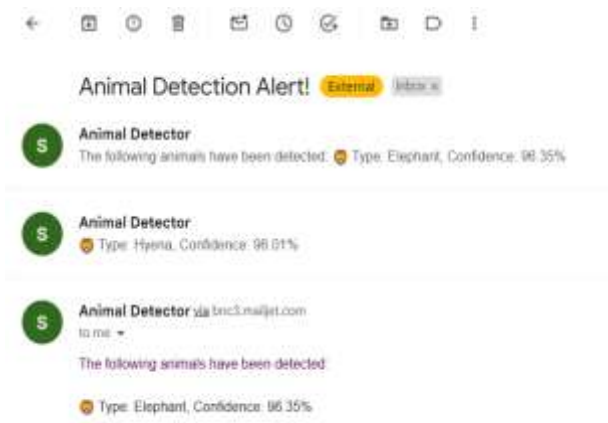


Figure 4- Email Notification



Figure 5- Animal Detection

This methodology facilitates real-time monitoring and alerting for animal intrusion by continuously analyzing input data from sensors and cameras. When an intrusion is detected, the system automatically generates and sends alerts via Email, notifying users or authorities about the detected animal along with relevant details such as detection confidence

IV. EXPERIMENTAL ANALYSIS AND RESULTS

Parameter	Percentage
mAP	85.6
Precision	83.4
Recall	78

The animal detection model was evaluated using standard performance metrics, including mean Average Precision (mAP), precision, and recall. Our model achieved an mAP of **85.6%**, demonstrating its strong overall detection capability. The precision of **83.4%** indicates a high proportion of correctly identified animals among the detected instances, reducing false positives. Additionally, the recall of **78.0%** reflects the model's ability to detect a significant proportion of actual animals in the dataset, minimizing false negatives. These results highlight the effectiveness of our approach in accurately identifying animal intrusions, making it suitable for real-world deployment in farm environments.

VI. CONCLUSION

The WildEcoWatch project leverages IoT and AI to provide an intelligent solution for wildlife conservation by detecting, monitoring, and managing animal intrusions in forests, farms, and conservation zones. The system integrates AI-based animal detection models and IoT sensors for real-time threat identification, sending instant alerts to forest rangers and law enforcement to prevent poaching, illegal logging, and habitat destruction.

Designed for sustainability, it operates in remote areas using solar power and energy-efficient microcontrollers while providing voice alerts and SMS notifications for timely interventions. In agriculture, it helps protect crops from wildlife intrusions, reducing losses and boosting productivity. With a scalable and cost-effective design, WildEcoWatch aims to enhance biodiversity conservation and human-wildlife coexistence, making it a vital tool for sustainable environmental protection.

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